

IDENTIFICATION

PRODUCT CODE:	MAINDEC-8E-D0DB-D
PRODUCT NAME:	RANDOM AND TEST
DATE CREATED:	JUNE 7, 1971
MAINTAINER:	DIAGNOSTIC GROUP
AUTHOR:	MICHAEL DAVIS

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1. ABSTRACT

THIS PROGRAM TESTS THE AND INSTRUCTING OF THE PDP-8E, THE AND INSTRUCTION, INSTRUCTION ADDRESS, OPERAND ADDRESS AND BOTH OPERANDS ARE PRODUCED BY RANDOM NUMBER GENERATORS,

2. REQUIREMENTS

2.1 EQUIPMENT

PDP-8E EQUIPPED WITH AT LEAST 4K OF MEMORY,

TELETYPE,

2.2 STORAGE

THE PROGRAM IS INITIALLY LOADED INTO LOCATIONS 0000 THRU 1177, THE INITIAL TEST AREA IS 1200-7777, WHEN THE PROGRAM RELOCATES, IT OCCUPIES 6600-7777, THE TEST AREA IS THEN 0000-6577,

2.3 PRELIMINARY PROGRAMS

MAINDEC-8E-D0AA, D0BA, D0CA

3. LOADING PROCEDURE

THE STANDARD PROCEDURE FOR LOADING BINARY TAPES IS TO BE USED.

4. STARTING PROCEDURE

4.1 STARTING ADDRESS

0200

4.2 CONTROL SWITCH SETTINGS

SR00=1, SUPPRESS HALT ON ERROR
SR01=1, HALT AT END OF PASS, RESTORE LOADERS
SR02=1, SUPPRESS PROGRAM RELOCATION
SR03=1, SUPPRESS END OF PASS TYPEOUT
SR09=1, HOLD DATA 1 CONSTANT
SR10=1, HOLD DATA 2 CONSTANT
SR11=1, HOLD INSTRUCTION CONSTANT

4.3 OPERATOR ACTION

4.3.1 SET SR TO 0200

4.3.2 PRESS LOAD ADDRESS SWITCH

4.3.3 SET SR TO 0000

4.3.4 PRESS CLEAR AND CONTINUE SWITCHES

5. OPERATING PROCEDURE

SAME AS 4.

6. ERRORS

6.1 RELOCATION ERROR

IF AN ERROR OCCURS DURING PROGRAM RELOCATION, THE PROGRAM WILL HALT AT 234 OR 6634, DEPENDING UPON WHETHER THE PROGRAM IS LOCATED LOW OR HIGH,

6.2 DATA ERRORS

IF THE LINK IS SET AFTER COMPLETION OF THE AND INSTRUCTION, OR IF THE RESULTS OF THE AND INSTRUCTION ARE INCORRECT, THE PROGRAM WILL HALT AT 731(7331) WITH DATA1 IN THE AC,

DEPRESS CONTINUE TO DISPLAY DATA2 IN THE AC,
DEPRESS CONTINUE TO DISPLAY AND INSTRUCTION IN AC,
DEPRESS CONTINUE TO DISPLAY INSTRUCTION ADDRESS IN AC,
DEPRESS CONTINUE TO DISPLAY DATA2 ADDRESS IN AC,
DEPRESS CONTINUE TO DISPLAY INDIRECT POINTER (USED BY INDIRECT AND) IN AC,
DEPRESS CONTINUE TO RESUME TEST

6.3 ERROR RECOVERY

6.3.1 RELOCATION ERROR

RELOAD PROGRAM

6.3.2 DATA ERROR

SEE 6.2

6.4 LOOPING

SET SR00=1 TO PREVENT HALT AFTER ERROR,
SET SR02=1 TO PREVENT RELOCATION,
SET SR03=1 TO SUPPRESS END OF PASS TYPEOUT,
SET SR09=SR11=1 TO HOLD INSTRUCTION AND DATA CONSTANT,

7. RESTRICTIONS

NONE

8. EXECUTION TIME

THE PROGRAM PERFORMS 4096 RANDOM TESTS IN APPROXIMATELY 2 SECONDS AND THEN RELOCATES. THE PROGRAM WILL TYPE "A" AFTER EACH 4096 RANDOM TESTS UNLESS SR03=1.

9. PROGRAM DESCRIPTION

THE PROGRAM IS INITIALLY LOADED INTO LOCATIONS 0200-1177, WITH TEMPORARY STORAGE LOCATIONS ON PAGE 0. AFTER INITIAL KEYSTART, THE PROGRAM WILL SAVE R1M AND 01N LOADERS IN PAGE 0 AND WILL THEN PROCEED TO USE LOCATIONS 1200-7777 AS A TEST AREA.

THE PROGRAM USES SEPARATE RANDOM NUMBER GENERATORS TO GENERATE THE AND INSTRUCTION, INSTRUCTION AND DATA ADDRESSES, AND THE TWO ARGUMENTS TO BE "AND'ED". THE INSTRUCTIONS AND DATA ARE STORED IN THEIR PREVIOUSLY GENERATED ADDRESSES. THE PROGRAM TRANSFERS TO THE LOCATION OF THE INSTRUCTION AND EXECUTES IT. THE PROGRAM THEN TRANSFERS TO A COMPARISON ROUTINE WHERE THE ACTUAL RESULT OF THE AND INSTRUCTION IS COMPARED TO A SIMULATED AND.

AFTER 4096 TESTS, THE PROGRAM TYPES "A", RELOCATES, AND CONTINUES TESTING.

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/
/RANDOM AND TEST
/COPYRIGHT 1970, DIGITAL EQUIPMENT CORP., MAYNARD, MASS. 01754
/V 82 07552
/
/
/TEMPORARY TRANSFER LOCATIONS ON PAGE 0
/
0000 0000 *0
0000 0000 TTANDL, 0
0001 5001 INSTL, JMP
0002 0002 INADDL, 2 /STORAGE FOR AND INSTRUCTION
0003 0003 DATADL, 3 /STORAGE FOR AND INSTRUCTION ADDRESS
0004 0003 IFLAGL, 3 /STORAGE FOR DATA ADDRESS
0005 0000 PADDL, 0 /STORAGE FOR INSTRUCTION INDIRECT FLAG
0006 0000 LIML, 0 /STORAGE FOR DATA INDIRECT ADDRESS
0007 0000 PAGL, 0

0010 0010 *10
0010 0000 PNTRL, 0

6007 CAF=6007
7501 MQA=7501
7421 MQL=7421
0266 CNTRL=LOHIL
0020 0020 *20
0020 0000 TEMP3L, 0

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/
/INITIALIZATION AND CONTROL ROUTINES
/
0200 0200 *200
0200 6007 STARTL, CAF
0201 1204 TAD .+3 /SAVE RIM AND BIN IN PAGE 0
0202 3201 PNTRL, DCA .-1 /ONLY AT INITIAL KEYSTART
0203 4205 PNTRL, JMS SAVBNL /201 BECOMES JMP SETLKL
0204 5274 TEMP1L, JMP SETLKL /FOR ALL FUTURE PASSES
/
/TRANSFER RIM AND BIN LOADERS TO PAGE 0
/
0205 0000 SAVBNL, 0
0206 1374 TAD C7600L /SET AC=-200, NUMBER OF
0207 3266 DCA CNTRL /LOCATIONS TO BE TRANSFERRED
0210 1374 TAD C7600L /FIRST "FROM"
0211 3202 DCA PNTRL /LOCATION=7600
0212 3203 DCA PNTRL /FIRST "TO" LOCATION=0
0213 4225 JMS RELOL /PERFORM TRANSFER
0214 5605 JMP I SAVBNL /EXIT

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0215 0000 RESBNL, 0
0216 1374 TAD C7600L /SET AC=-200, NUMBER OF
0217 3266 DCA CNTR1L /LOCATIONS TO BE TRANSFERRED
0220 3202 DCA PNTR1L /FIRST "FROM" ADDRESS=0
0221 1374 TAD C7600L /FIRST "TO"
0222 3203 DCA PNTR2L /ADDRESS=7600
0223 4225 JMS RELOL /PERFORM TRANSFER
0224 5615 JMP I RESBNL /EXIT

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/

/DATA TRANSFER ROUTINE

/

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0225 0000 RELOL, 0
0226 1602 TAD I PNTR1L /GET DATA
0227 3603 DCA I PNTR2L /TRANSFER
0230 1602 TAD I PNTR1L /GET DATA
0231 7041 CIA
0232 1603 TAD I PNTR2L /COMPARE
0233 7640 SZA CLA
0234 7402 WLT /TRANSFER ERROR
0235 2202 ISZ PNTR1L /NEXT "FROM" LOCATION
0236 7000 NOP
0237 2203 ISZ PNTR2L /NEXT "TO" LOCATION
0240 7000 NOP
0241 2266 ISZ CNTR1L
0242 5226 JMP RELOL+1
0243 5625 JMP I RELOL

```

/

/TRANSFER PROGRAM TO UPPER MEMORY

/

```

0244 4215 REHL, JMS RESBNL /TRANSFER RIM AND BIN LOADERS
0245 1373 TAD C7000L /SET AC=-1000, NUMBER OF
0246 3266 DCA CNTR1L /LOCATIONS TO BE TRANSFERRED
0247 1365 TAD C200L /FIRST "FROM"
0250 3202 DCA PNTR1L /ADDRESS=200
0251 1372 TAD C6600L /FIRST "TO"
0252 3203 DCA PNTR2L /ADDRESS=6600
0253 4225 JMS RELOL /PERFORM TRANSFER
0254 5772 JMP I C6600L /GO TO PROGRAM START

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/

/TRANSFER PROGRAM TO LOWER MEMORY

/

```

0255 4205 REL, JMS SAVBNL /TRANSFER RIM AND BIN LOADERS
0256 1373 TAD C7000L /SET AC=-1000, NUMBER OF
0257 3266 DCA CNTR1L /LOCATIONS TO BE TRANSFERRED
0260 1372 TAD C6600L /FIRST "FROM"
0261 3202 DCA PNTR1L /ADDRESS=6600
0262 1365 TAD C200L /FIRST "TO"
0263 3203 DCA PNTR2L /ADDRESS=200

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0264 4225 JMS RELOL /PERFORM TRANSFER
0265 5765 JMP I C200L /GO TO PROGRAM START

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/
/ DETERMINE IF PROGRAM IS IN LOWER OR UPPER MEMORY
/
0266 0000 LOHIL, 0 /PC
0267 7330 CLA CLL CML RAR /SET AC=4000
0270 1266 TAD -2 /ADD PC
0271 7630 SZL CLA /IS LINK=0
0272 1371 TAD C6400L /NO, HIGH CORE
0273 5666 JMP I LOHIL /RETURN
/
/ TRANSFER TO LINKAGE GENERATION
/
0274 7300 SETLKL, CLA CLL
0275 3202 DCA PNTR1L /CLEAR PASS COUNTER
0276 4266 JMS LOHIL /DETERMINE IF PROGRAM IS HIGH OR LOW
0277 5377 JMP GOSETL /GO TO TEST WITH ADDRESS MODIFIER IN AC
/
/ SIMULATE LOGICAL AND WITH A AND B=NOT((NOTA)OR(NOTB))
/
0300 0000 SANDL, 0
0301 7040 CMA /NOTA
0302 3204 DCA TEMP1L /SAVE NOTA
0303 7501 MQA /GET B
0304 7040 CMA /NOTB
0305 7421 MQL /SAVE NOTB
0306 1204 TAD TEMP1L /GET NOTA
0307 7501 MQA /OR WITH NOTB
0310 7040 CMA /COMPLEMENT
0311 5700 JMP I SANDL /EXIT WITH RESULT IN AC
/
/ TEST SWITCHES
/
0312 0000 SWITCL, 0
0313 7421 MQL /SAVE TEST BIT
0314 7604 LAG /GET SWITCHES
0315 4300 JMS SANDL /AND SWITCHES WITH TEST BIT
0316 7650 SNA CLA /IS SWITCH 0
0317 2312 ISE SWITCL /NO SKIP INSTRUCTION AFTER RETURN
0320 5712 JMP I SWITCL

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/
/ END OF PASS
/
0321 0000 EPASL, 0
0322 2202 ISE PNTR1L /END OF PASS ?
0323 5721 JMP I EPASL /NO, RETURN
0324 7332 CLA CLL CML RTR
0325 7012 RTR /SET AC=4000
0326 4312 JMS SWITCL /TEST SR03
0327 5336 JMP ,+7 /SUPPRESS END OF PASS TYPEOUT
0330 1366 TAD C215L
0331 4337 JMS TYPEL /TYPE CARRIAGE RETURN
0332 1367 TAD C212L
0333 4337 JMS TYPEL /TYPE LINEFEED
0334 1370 TAD A
0335 4337 JMS TYPEL /TYPE

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0336 5345      JMP      HALTL      /TEST FOR HALT, RELOCATION
/
/OUTPUT CHARACTER
/
0337 0000      TYPEL, 0
0340 6046      TLS
0341 6041      TSF
0342 5341      JMP      , -1
0343 7200      CLA
0344 5737      JMP I    TYPEL

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/
/CHECK FOR HALT
/
0345 7332      HALTL, CLA CLL CML RTR      /SET AC=2000
0346 4312      JMS      SWITCL      /TEST SR01
0347 7410      SKP
0350 5355      JMP      RRELL      /SR01=1, HALT
0351 4266      JMS      LOHIL      /CHECK FOR RELOCATION
0352 7650      SNA CLA      /DETERMINE IF PROGRAM IS HIGH OR LOW
0353 4215      JMS      RESBNL      /AC=0, PROGRAM LOW
0354 7402      HLT      /PROGRAM LOW, RESTORE LOADERS

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/
/CHECK FOR RELOCATION
/
0355 7332      RRELL, CLA CLL CML RTR
0356 7010      RAR      /SET AC=1000
0357 4312      JMS      SWITCL      /TEST SR02
0360 5721      JMP I    EPASL      /SR02=1, DO NOT RELOCATE PROGRAM
0361 4266      JMS      LOHIL      /DETERMINE IF PROGRAM IS HIGH OR LOW
0362 7650      SNA CLA      /AC=0, PROGRAM LOW
0363 5244      JMP      REHL      /PROGRAM LOW, RELOCATE TO HIGH CORE
0364 5255      JMP      REL      /PROGRAM HIGH, RELOCATE TO LOW CORE
0365 0200      C200L, 200
0366 0215      C215L, 215
0367 0212      C212L, 212
0370 0301      A,      301
0371 6400      C6400L, 6400
0372 6600      C6600L, 6600
0373 7000      C7000L, 7000
0374 7600      C7600L, 7600

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0377 0377      *377
0377 7000      GOSCTL, NOP

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/
/SET UP ADDRESS POINTERS AND CONSTANTS AND TRANSFER TO NEXT PAGE
/
0400 3237      *400      SETAL, DCA      TEMPL      /SAVE ADDRESS MODIFIER
0401 1242      TAD      LISTL      /GET POINTER FOR TRANSFER
0402 1237      TAD      TEMPL      /MODIFY FOR LOW OR HIGH CORE

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0403	3010	DCA	PNTRL	/SET UP AUTO-INDEX REGISTER
0404	1243	TAD	LGENL	/GET POINTER TO INSTRUCTION GENERATION
0405	1237	TAD	TEMPL	/MODIFY FOR LOW OR HIGH CORE
0406	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0407	1245	TAD	LSANDL	/GET POINTER TO AND SIMULATOR
0410	1237	TAD	TEMPL	/MODIFY FOR LOW OR HIGH CORE
0411	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0412	1246	TAD	LRETUL	/GET POINTER FOR EXECUTION RETURN
0413	1237	TAD	TEMPL	/MODIFY FOR LOW OR HIGH CORE
0414	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0415	1247	TAD	LSWITL	/GET POINTER TO SWITCH SENSING
0416	1237	TAD	TEMPL	/MODIFY FOR LOW OR HIGH CORE
0417	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0420	1244	TAD	LEPASL	/GET POINTER TO END OF PASS
0421	1237	TAD	TEMPL	/MODIFY FOR LOW OR HIGH CORE
0422	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0423	1237	TAD	TEMPL	/GET ADDRESS MODIFIER
0424	7640	SZA CLA		/IS TEST IN LOW CORE
0425	5233	JMP	LHICOL	/NO,SET UP FOR HIGH CORE
0426	1240	TAD	L200L	/SET PAGE 0 EXCLUSION BIT
0427	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0430	1250	TAD	L6600L	/GET LOW CORE ADDRESS LIMIT
0431	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0432	5377	JMP	GOTSTL	/GO TO TEST
0433	3410	LHICOL, DCA I	PNTRL	/CLERA PAGE 0 EXCLUSION BIT
0434	1251	TAD	L1201L	/GET HIGH CORE ADDRESS LIMIT
0435	3410	DCA I	PNTRL	/TRANSFER TO NEXT PAGE
0436	5377	JMP	GOTSTL	/GO TO TEST

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0437	0000	TEMPL, 0
0440	0200	L200L, 200
0441	7000	L7000L, 7000
0442	0753	LISTL, TGENL-1
0443	1000	LGENL, GENL
0444	0521	LEPASL, EPASL
0445	0300	LSANDL, SANDL
0446	0712	LRETUL, TRETUL
0447	0312	LSWITL, SWITCL
0450	6600	L6600L, 6600
0451	1201	L1201L, 1201
	0577	*577
0577	7000	GOTSTL, NOP

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/

/GENERATE TEST INSTRUCTION AND DATA

/

0600	0600	*600		
0600	7300	TEST1L, CLA CLL		
0601	1355	TAD	TANDL	/GET POINTER TO SIMULATED AND
0602	3000	DCA	TTANDL	/PLACE IN TRANSFER LOCATION
0603	7001	IAC		/SET
0604	4757	JMS I	TSWITL	/TEST 21

0605	5224	JMP	TDAT1L	/SR	DO NOT GENERATE INSTRUCTION
0606	1362	TAD	TLIML	/NO	ADDRESS LIMIT
0607	3026	DCA	LIML	/SAVE	
0610	1361	TAD	TPAGBL	/GET	PAGE EXCLUSION BIT
0611	3027	DCA	PAGL		
0612	4754	JMS I	TGENL	/GENERATE	INSTRUCTION
0613	3365	DCA	TIFLGL	/SAVE	INDIRECT FLA
0614	1001	TAD	INSTL	/GET	INSTRUCTION
0615	3363	DCA	TINSTL	/SAVE	IT
0616	1002	TAD	INADDL	/GET	INSTRUCTION ADDRESS
0617	3364	DCA	TINADL	/SAVE	IT
0620	1003	TAD	DATADL	/GET	DATA ADDRESS
0621	3366	DCA	TDATAL	/SAVE	IT
0622	1005	TAD	PADDL	/GET	INDIRECT TD DATA
0623	3367	DCA	TPADDL	/SAVE	IT
0624	7105	TDAT1L, CLL IAC	RAL	/SET	AC=2
0625	4757	JMS I	TSWITL	/TEST	SR10
0626	5234	JMP	TDAT2L	/SR10=1, DO NOT	GENERATE DATA1
0627	1370	TAD	TDA1L	/GENERATE	RANDOM NUMBER
0630	7104	CLL RAL			
0631	7430	SZL			
0632	1374	TAD	T3L		
0633	3370	DCA	TDA1L		
0634	7307	TDAT2L, CLA CLL IAC RTL		/SET	AC=4
0635	4757	JMS I	TSWITL	/TEST	SR09
0636	5244	JMP	SETTL	/SR09=1, DO NOT	GENERATE DATA2
0637	1371	TAD	TDA2L	/GENERATE	RANDOM NUMBER
0640	7104	CLL RAL			
0641	7430	SZL			
0642	1374	TAD	T3L		
0643	3371	DCA	TDA2L		
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/
/SET UP INSTRUCTION AND DATA AT TEST ADDRESS
/ALONG WITH RETURN TO THIS ROUTINE
/
0644 7300 SETTLL, CLA CLL
0645 1363 TAD TINSTL /GET INSTRUCTION
0646 3764 DCA I TINADL /STORE IN TEST LOCATION
0647 1365 TAD TIFLGL /GET INDIRECT FLAG
0650 7650 SNA CLA /IS INSTRUCTION INDIRECT
0651 5267 JMP DURL /NO, GET DATA
0652 1366 TAD TDATAL /INDIRECT, IS ADDRESS
0653 1375 TAD T7760 /AUTO-INDEX REGISTER
0654 7630 SZL CLA
0655 5262 JMP .+5 /NO, USE POINTER AS IS
0656 1366 TAD TDATAL
0657 1376 TAD T7770
0660 7630 SZL CLA
0661 7040 CMA /ADDRESS IS AUTO-INDEX REGISTER
0662 1367 TAD TPADDL /GET INDIRECT ADDRESS
0663 3766 DCA I TDATAL /STORE IN TEST LOCATION
0664 1370 TAD TDA1L /GET DATA
0665 3767 DCA I TPADDL /STORE IN TEST LOCATION
0666 5271 JMP DOTSTL
0667 1370 DURL, TAD TDA1L /GET DATA

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0670	3766	DCA I	TOTAL	/STORE IN TEST LOCATION
		/		
		/SIMULATE "AND"		
		/		
0671	7300	DOTSTL	CLA CLL	
0672	1370	TAD	TDA1L	/GET DATA1
0673	7421	MQL		/SAVE IN MQ
0674	1371	TAD	TDA2L	/GET DATA2
0675	4755	JMS I	TANDL	/DO SIMULATION
0676	3372	DCA	TSIML	/SAVE ANSWER
		/		
		/GO TO TEST		
		/		
0677	1356	DOANDL	TAD	TRETTL
0700	3000	DCA	TTANDL	/GET RETURN ADDRESS
0701	1364	TAD	TINADL	/SAVE
0702	7001	IAC		/GET INSTRUCTION ADDRESS
0703	7450	SNA		/INCREMENT
0704	5200	JMP	TEST1L	/IS IT 0
0705	3353	DCA	TEMP2L	/YES, GENERATE NEW INFORMATION
0706	1373	TAD	T5400L	/NO, SAVE
0707	3753	DCA I	TEMP2L	/GET RETURN INSTRUCTION
0710	1371	TAD	TDA2L	/PUT IN TEST LOCATION
0711	5764	JMP I	TINADL	/GET DATA2
				/EXECUTE "AND"

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		/		
		/RETURN HERE AFTER EXECUTION		
		/		
0712	3377	TRETUL	DCA	TRACL
0713	7430	SEL		/SAVE AC
0714	4324	JMS	ERROR	/IS LINK=1
0715	1372	TAD	TSIML	/LINK=1, ERROR
0716	7041	CLA		/GET SIMULATION RESULT
0717	1377	TAD	TRACL	
0720	7640	SZA CLA		/ADD REAL RESULT
0721	4324	JMS	ERROR	/ARE THEY THE SAME
0722	4760	JMS I	TEPASL	/NO, ERROR
0723	5200	JMP	TEST1L	/END OF PASS
		/		
		/ERROR HANDLER		
		/		
0724	0000	ERROR,	0	
0725	7330	CLA CLL	CML RAR	/SET AC=4000
0726	4757	JMS I	TSWITL	/TEST SR00
0727	5351	JMP	TEXTIL	/SR00=1, DO NOT HALT ON ERROR
0730	1370	TAD	TDA1L	/DISPLAY DATA1 IN AC
0731	7402	HLT		
0732	7200	CLA		
0733	1371	TAD	TDA2L	/DISPLAY DATA2 IN AC
0734	7402	HLT		
0735	7200	CLA		
0736	1001	TAD	INSTL	/DISPLAY INSTRUCTION IN AC
0737	7402	HLT		
740	7200	CLA		
741	1364	TAD	TINADL	/DISPL) INSTRUCTION ADDRESS IN AC

0742	7402	HLT		
0743	7200	CLA		
0744	1366	TAD	TOTAL	/DSI, LAY DATA ADDRESS IN AC
0745	7402	HLT		
0746	7200	CLA		
0747	1367	TAD	TPADDL	/DSIPLAY INDIRECT IN AC
0750	7402	HLT		
0751	7300	TEXTL, CLA CLL		
0752	5724	JMP I	ERROR	

0753	0000	TEMP2L, 0
0754	0000	TGENL, 0
0755	0000	TANDL, 0
0756	0000	TRETTL, 0
0757	0000	TSWITL, 0
0760	0000	TEPASL, 0
0761	0000	TPAGBL, 0
0762	0000	TLIML, 0
0763	0000	TINSTL, 0
0764	0000	TINADL, 0
0765	0000	TIFLGL, 0

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0766	0000	TOTAL, 0
0767	0000	TPADDL, 0
0770	0021	TOA1L, 21
0771	0037	TOA2L, 37
0772	0000	TSIML, 0
0773	5400	T5400L, 5400
0774	0003	T3L, 3
0775	7760	T7760, 7760
0776	7770	T7770, 7770
0777	0000	TRACL, 0

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/

/GENERATE INSTRUCTIONS AND ADDRESSES

/

1000	1000	*1000
0000	0000	GENL, 0

/

/GENERATE "AND" INSTRUCTION

/

1001	1367	GANDL, TAD	R1L	
1002	4340	JMS	RANDL	/GENERATE RANDOM NUMBER
1003	3367	DCA	R1L	/SAVE NUMBER
1004	1367	TAD	R1L	
1005	7421	SQL		
1006	1007	TAD	PAGL	
1007	7501	MQA		/OR RANDOM NUMBER WITH EXCLUSION BIT
1010	7421	SQL		
1011	1371	TAD	K0777	/MASK OFF 3MSB
1012	4400	JMS I	TTANDL	/TO GET "0" OP CODE
1013	3001	DCA	INSTL	/SAVE INSTRUCTION
1014	1001	TAD	INSTL	/GET INSTRUCTION

1015	4354	JMS	AND17L	/EXTRACT PAGE ADDRESS OF INSTRUCTION
1016	3020	OCA	TEMP3L	/SAVE PAGE ADDRESS OF INSTRUCTION
/				
/GENERATE ADDRESS FOR INSTRUCTION				
/				
1017	1372	GENAOL,	TAD R2L	
1020	4340	JMS	RANDL	/GENERATE RANOOM NUMBER
1021	3372	DCA	R2L	/SAVE NUMBER
1022	4345	JMS	CLIML	/SET UP TO TEST ADDRESS LIMITS
1023	1372	TAD	R2L	
1024	7620	SNL CLA		/IS ADDRESS WITHIN LIMITS
1025	5217	JMP	GANADL	/NO, GENERATE NEW ADDRESS
1026	1372	TAD	R2L	
1027	1373	TAD	P0L	
1030	7620	SNL CLA		/IS ADDRESS ON PAGE 0
1031	5246	JMP	PAGADL	/NO
1032	1020	TAD	TEMP3L	/GET PAGE ADDRESS OF INSTRUCTION
1033	7041	CIA		
1034	1372	TAD	R2L	/SUBTRACT ADDRESS
1035	4361	PAGAL,	JMS ABSL	
1036	7700	SMA CLA		/IS DIFFERENCE >2
1037	5217	JMP	GANADL	/NO
1040	1020	PAL,	TAD TEMP3L	/GET PAGE ADDRESS OF INSTRUCTION
1041	7650	SNA CLA		/DOES INSTRUCTION REFERENCE LOCATION 0
1042	5201	JMP	GANDL	/YES, GENERATE NEW INSTRUCTION
1043	1372	TAD	R2L	/YES, USE ADDRESS
1044	3002	DCA	INADDL	
1045	5261	JMP	DAADL	/GENERATE ADDRESS FOR DATA
1046	1001	PAGADL,	TAD INSTL	/GET INSTRUCTION
1047	7421	MQL		
1050	1376	TAD	K200L	
1051	4400	JMS I	TTANDL	/MASK CURRENT PAGE BIT
1052	7650	SNA CLA		/IS PAGE BIT SET
1053	5240	JMP	PAL	/NO, USE ADDRESS AS IS

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1054	1372	TAD	R2L
1055	4354	JMS	AND17L
1056	7041	CIA	
1057	1020	TAD	TEMP3L
1060	5235	JMP	PAGAL

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/

/GENERATE ADDRESS FOR DATA

/

1061	1001	DAADL,	TAD INSTL	/GET INSTRUCTION
1062	7421	MQL		
1063	1376	TAD	K200L	
1064	4400	JMS I	TTANDL	/DOES INSTRUCTION REFERENCE PAGE 0
1065	7650	SNA CLA		
1066	5306	JMP	P0AL	/YES
1067	1002	TAD	INADDL	
1070	7421	MQL		
1071	1373	TAD	P0L	

1073	7421	JMS I	TTANDL	/EXIT	1 PAGE OF ADDRESS
1074	1020	MQL			
1075	7501	TAD	TEMP3L		
1076	3003	HQA		/OR" TOGETHER TO GET	
1077	1001	DCA	DATADL	/DATA ADDRESS	
1100	7421	INDIRL, TAD	INSTL		
1101	1375	MQL			
1102	4400	TAD	K400L		
1103	7640	JMS I	TTANDL		
1104	5311	SZA CLA		/IS INSTRUCTION INDIRECT	
1105	5600	JMP	PADL	/YES, INSTRUCTION IS INDIRECT	
1106	1020	JMP I	GENL	/EXIT	
1107	3003	P0AL, TAD	TEMP3L		
1110	5277	DCA	DATADL		
		JMP	INDIRL		
		/			
		/GENERATE INDIRECT ADDRESS FOR DATA			
		/			
1111	1377	PADL, TAD	R3L	/GENERATE RANDOM NUMBER	
1112	4340	JMS	RANDL		
1113	3377	DCA	R3L		
1114	4345	JMS	CLIML		
1115	1377	TAD	R3L		
1116	7620	SNL CLA		/IS ADDRESS WITHIN LIMITS	
1117	5311	JMP	PADL	/NO, TRY AGAIN	
1120	1002	TAD	INADDL	/GET INSTRUCTION ADDRESS	
1121	7041	CIA			
1122	1377	TAD	R3L	/SUBTRACT INDIRECT	
1123	4361	JMS	ABSL	/GENERATE -ABSOLUTE VALUE	
1124	7700	SMA CLA		/DO INSTRUCTION AND ADDRESS INTERFERE	
1125	5311	JMP	PADL	/YES	
1126	1003	TAD	DATADL		
1127	7041	CIA			
1130	1377	TAD	R3L		
1131	4361	JMS	ABSL		
1132	7700	SMA CLA			
1133	5311	JMP	PADL		
1134	1377	TAD	R3L	/NO	
1135	3005	DCA	PADDL		
1136	7040	CMA			
/	PAL10	V141	17-JUN-71	7:23	PAGE 12-1
1137	5600	JMP I	GENL	/EXIT	
/	PAL10	V141	17-JUN-71	7:23	PAGE 13
1140	0000	RANDL, 0			
1141	7104	CLL RAL			
1142	7430	SZL			
1143	1370	TAD	K3L		
1144	5740	JMP I	RANDL		
		/			
		/			
		/			
1145	0000	CLIML, 0			
1146	1007	TAD	PAGL		
1147	7100	CLL			

1150	7650	SNA CLA	
1151	7020		CML
1152	1006		TAD
1153	5745		JMP I CLIML
			/
			/
			/
1154	0000	AND17L,	0
1155	7421		MQL
1156	1374		TAD K0177L
1157	4400		JMS I TTANDL
1160	5754		JMP I AND17L
			/
			/
			/
1161	0000	ABSL,	0
1162	7500		SMA
1163	7041		CIA
1164	7001		IAC
1165	7001		IAC
1166	5761		JMP I ABSL
			/
			/
			/
1167	0001	R1L,	1
1170	0003	K3L,	3
1171	0777	K0777,	777
1172	0005	R2L,	5
1173	7600	P0L,	7600
1174	0177	K0177L,	177
1175	0400	K400L,	400
1176	0200	K200L,	200
1177	0015	R3L,	15

S

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0000	11111111	10000000	10000000	00000000	00000000	00000000	00000000	00000000
0100	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111001
0400	11111111	11111111	11111111	11111111	11111111	11000000	00000000	00000000
0500	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000001
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1200								
1300								

1400
1500

1600
1700

2000
2100

2200
2300

2400
2500

2600
2700

3000
3100

3200
3300

3400
3500

3600
3700

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4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

/	PAL10	V141	17-JUN-71	7123	PAGE 13-3
A	0370		PADDL	0005	TSIML 0772
ABSL	1161		PADL	1111	TSWITL 0757
AND17L	1154		PAGADL	1046	TTANDL 0000
C200L	0365		PAGAL	1035	TYPEL 0337
C212L	0367		PAGL	0007	
C215L	0366		PAL	1040	
C6400L	0371		PNTR1L	0202	
C6600L	0372		PNTR2L	0203	
C7000L	0373		PNTRL	0010	
C7600L	0374		R1L	1167	
CAF	6007		R2L	1172	
CLIML	1145		R3L	1177	
CNTR1L	0266		RANDL	1140	
DAADL	1061		REHL	0244	
DATADL	0003		RELL	0255	
DIRL	0667		RELOL	0225	
DOANDL	0677		RESBNL	0215	
DOTSTL	0671		RRELL	0355	
EPASL	0321		SANDL	0300	
ERROR	0724		SAVBNL	0205	
GANADL	1017		SETAL	0400	
GANDL	1001		SETLKL	0274	
GENL	1000		SETTL	0644	
GOSETL	0377		STARTL	0200	
GOTSTL	0577		SWITCL	0312	
HALTL	0345		T3L	0774	
IFLAGL	0004		T5400L	0773	
INADDL	0002		T7760	0775	
INDIRL	1077		T7770	0776	
INSTL	0001		TANDL	0755	
)77L	1174		TDA1L	0770	
77	1171		TDA2L	0771	

K7 1176
 KL 1170
 K400L 1175
 L1201L 0451
 L200L 0440
 L6600L 0450
 L7000L 0441
 LEPASL 0444
 LGENL 0443
 LHICOL 0433
 LIML 0006
 LISTL 0442
 LOHIL 0266
 LRETUL 0446
 LSANDL 0445
 LSWITL 0447
 MQA 7501
 MQL 7421
 P0AL 1106
 P0L 1173

T0AT1L 0624
 T0AT2L 0634
 T0ATAL 0766
 TEMP1L 0204
 TEMP2L 0753
 TEMP3L 0020
 TEMPL 0437
 TEPASL 0760
 TEST1L 0600
 TEXT1L 0751
 TGENL 0754
 TIFLGL 0765
 TINADL 0764
 TINSTL 0763
 TLIML 0762
 TPADDL 0767
 TPAGBL 0761
 TRACL 0777
 TRETTL 0756
 TRETUL 0712

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ERRORS DETECTED: 0

LINKS GENERATED: 0

RUN-TIME: 5 SECONDS

2K CORE USED